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Preventive effects of North American ginseng (*Panax quinquefolium*) on diabetic nephropathy

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ABSTRACT

Purpose: Ginseng has been used as an herbal medicine and nutritional supplement in East Asia for thousands of years and gained popularity in the west because of its various pharmacological properties. *Panax ginseng* (Asian ginseng) and *Panax quinquefolium* (North American ginseng) both are reported to possess antihyperglycemic properties. The aim of the present study is to evaluate the preventive effects of North American ginseng on diabetic nephropathy (DN) and the underlying mechanisms of such effects.

Methods: Models of both type 1 (C57BL/6 mice with STZ-induced diabetes) and type 2 diabetes (db/db mice) and age- and sex-matched controls were examined. Alcoholic ginseng root (200 mg/kg body wt, daily oral gavage) extract was administered to the diabetic mice (type 1 and type 2) for two or four months in order to evaluate its effects on DN.

Results: Dysmetabolic state in the diabetic mice was significantly improved by ginseng treatment. In the kidneys of diabetic animals, ginseng significantly prevented oxidative stress and reduced the NF-κB (p65) levels. Diabetes-induced up-regulations of ECM proteins and vasoactive factors in the kidneys were significantly diminished by ginseng administration. Furthermore, albuminuria and mesangial expansion in the diabetic mice were prevented by ginseng therapy.

Conclusion: North American ginseng has preventive effects on DN and it works through a combination of mechanisms such as antihyperglycemic and antioxidant activities.

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Introduction

Diabetes mellitus, characterized by hyperglycemia and long-term complications is the most common endocrine disorder. Such complications affect the kidneys (nephropathy), eyes (retinopathy), heart (cardiomyopathy), nerves (neuropathy) and blood vessels. Among the chronic complications, diabetic nephropathy is a major cause of morbidity among diabetic patients. The structural hallmark of this disease is glomerular mesangial matrix expansion

Abbreviations: ECM, extracellular matrix protein; NF-κB, nuclear factor-κB; OGGA, Ontario ginseng growers association; STZ, streptozotocin; CO, controls; DM, type 1 diabetic mice; DM-GIN, type 1 diabetic mice treated with ginseng; db, type 2 diabetic mice; db2, type 2 diabetic mice; db-GIN, type 2 diabetic mice treated with ginseng; HOMA-IR, homeostasis model assessment for insulin resistance; AP-1, activator protein; DMSO-TBAC, dimethyl sulfoxide-tetrabutylammonium chloride; PMSF, phenyl methyl sulphonyl fluoride; HO-1, heme oxygenase-1; FN, fibronectin; EDB* FN, extracellular domain B positive splice variant of fibronectin; VEGF, vascular endothelial growth factor; ET-1, endothelin-1; TGF-β1, transforming growth factor-β1; PAS, periodic acid-schiff stain.

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(Nathan 1996; Steffes et al. 1989). Biochemically, such lesions are characterized by increased production of extracellular matrix proteins (Skena and Gesualdo 2005). Hyperglycemia is the key initiating factor in the development of all chronic diabetic complications including diabetic nephropathy (DCCT 1993; UKPDS 1998). It has been hypothesized that an increase in oxidative stress, as a result of chronic hyperglycemia, activates several signaling pathways that alter gene expression (Brownlee 2001). There are evidences that high glucose in diabetes mellitus activates multiple pathways that lead to the increased generation of reactive oxygen species (ROS). Such mechanisms include protein kinase C activation, non-enzymatic glycation, augmented polyol and hexosamine pathways (Ishii et al. 1998a,b; Howard et al. 2002). These alterations lead to increased vasoactive factor and ECM protein production. Increased ROS can cause cell abnormalities, reacting directly with nitric oxide (NO) to produce cytotoxic peroxynitrite (Bonnefont-Rousselot 2002; Stadler et al. 2003). ROS also stimulates activation of various transcription factors, such as nuclear factor-κB (NF-κB), which are involved in the regulation of inflammatory mediators (Scheck et al. 1992). We have previously demonstrated augmented ECM protein fibronectin (FN) production through activation of transcription factors such as NF-κB and AP-1 (activator protein) in diabetic nephropathy (Chen et al. 2003a).

NF- κ B activation further triggers the expression of TGF- β 1, which in turn also induces excessive ECM protein production (Ohga et al. 2007).

Ginseng, a traditional medicinal plant is gaining popularity in the west in the recent two decades (Xie et al. 2005; Lee 1992) for its various pharmacological properties. Ginsenosides are the major components of this herb. Structurally, ginsenosides are divided in protopanaxadiol ginsenosides and protopanaxatriol ginsenosides (Wang et al. 2008). The derivatives of protopanaxatriol include Rg1, Rg2, Re, and Rf and that of protopanaxadiol consists of Rb1, Rb2, Rc, and Rd (WHO Monographs 1999). Ginseng quenches free radicals, protects low density lipoproteins from oxidation and inhibits lipid peroxidation (Kim et al. 2002; Kim and Park 2003; Jung et al. 2005). In recent years, accumulating in vitro and in vivo evidences have indicated that ginseng possesses significant hypoglycemic activities (Dey et al. 2003; Liu et al. 2005; Luo and Luo 2006). Park et al. (2010) reported that the protopanaxadiol derivatives of ginseng caused significant prevention of glucose-induced fibronectin upregulation in primary cultured rat mesangial cells.

We have previously demonstrated prevention of glucose-induced increased vasoactive factors and ECM protein production in the endothelial cells by American ginseng extract (Sen et al. 2011). In the present communication, we investigated the preventive role of North American ginseng on diabetic nephropathy using models of type 1 and type 2 diabetes. We further elucidated the possible underlying mechanisms of such prevention.

Materials and methods

Reagents

All reagents were obtained from Sigma–Aldrich (Oakville, ON, Canada) unless otherwise specified.

Preparation of ginseng root extract

Four-year-old North American ginseng roots collected in 2007 from 5 different farms in Ontario, Canada were provided by the Ontario Ginseng Growers Association (OGGA). Dried ginseng root samples were shipped to Naturex (375 Huyler St. South Hackensack, NJ 07606, USA) for extraction. Ginseng extracts from each farm were prepared individually and they were combined to produce composite extracts, which were used for the present study. A detailed methodology for alcoholic extraction has been previously published (Sen et al. 2011; Azike et al. 2011). The concentrates were lyophilized with a freeze drier at -50°C under reduced pressure to produce ginseng alcoholic extract in powdered form. HPLC analysis of the alcoholic extract and the content of ginsenosides have been previously described (Sen et al. 2011; Azike et al. 2011).

Animals

All animals were cared for according to the Guiding Principle in the Care and Use of Animals. All experiments were approved by the University of Western Ontario council on animal care committee [Protocol (AUP)-2010-001]. Male C57BL/6 mice weighing 23–36 g were obtained from Charles River (Montreal, QC, Canada). To create a model of type 1 diabetes, groups of animals received 3 intraperitoneal injections of STZ (50 mg/kg in citrate buffer, pH 5.6) on alternate days (Feng et al. 2010). Controls were injected with the same volume of citrate buffer. Randomly selected diabetic animals were monitored for two months (DM1) or four months (DM2). Furthermore, groups of the diabetic mice were subjected to ginseng treatment for two (DM1-GIN) or four months (DM2-GIN). Alcoholic ginseng extract (200 mg/kg BW) was dissolved in Dulbecco's Phosphate Buffered Saline, 1X, pH: 7.4 and administered by

daily gavage. The type 1 diabetic mice were implanted with insulin pellets that released small doses (0.5 units) of insulin to prevent ketonuria (1 U/day, Linshin Canada Inc., Toronto, ON, Canada).

For type 2 diabetic model, male db/db (Lepr^{db}, DBA/J) mice and controls (32–38 gms.) were purchased from Jackson Laboratories, USA. Randomly selected diabetic animals were monitored for two months (db1) or four months (db2). Furthermore, groups of the diabetic mice were subjected to ginseng treatment for two (db1-GIN) or four months (db2-GIN). Alcoholic ginseng extract (200 mg/kg BW) was dissolved in Dulbecco's Phosphate Buffered Saline, 1X, pH: 7.4 and administered by daily gavage (Shin et al. 2010; Lee et al. 1987; Hwang et al. 2010).

The animals were sacrificed after the follow-up period. Portion of renal cortical tissues were snap-frozen. The remaining was fixed in 10% neutral-buffered formalin for paraffin embedding and histological analyses.

Measurement of metabolic parameters

The animals were monitored regularly for assessment of body weight and blood glucose (Glucometer, Free Style Freedom Lite Inc.). Glycated hemoglobin (GlycoHb) was measured by using the GlycoHb kit (Stanbio Laboratories, Texas, USA).

Measurement of plasma insulin

Mouse insulin ELISA kit (DRG Diagnostics, Germany) was used to measure the plasma insulin concentration. Insulin resistance in the db/db mice was determined by the homeostasis model assessment (HOMA) method using the following equation: HOMA value for insulin resistance (HOMA-IR) = fasting insulin (mU/l) $\times$ fasting glucose (mmol/l)/22.5 (Matthews et al. 1985).

Measurement of renal parameters

Urinary albumin levels were measured after 2 and 4 months of the onset of diabetes by ELISA using microalbumin estimation kit (Albuwell, Philadelphia, USA). Serum creatinine level was estimated by standard alkaline picrate method using the creatinine estimation kit (DetectX, Ann Arbor, MI, USA).

Estimation of superoxide anion ($\text{O}_2^{\bullet-}$) generation

Superoxide anion was measured in kidney tissues after extraction with dimethyl sulfoxide–tetrabutylammonium chloride (DMSO–TBAC) solution. TBAC– $\text{O}_2^{\bullet-}$ complex was then detected by use of the luminol–EDTA–Fe enhanced chemiluminescence method (Yao et al. 2004).

Nuclear protein extraction and NF- κ B assay

Nuclear protein was isolated from the kidneys as described elsewhere, with some modifications (Lee et al. 1998; Zhou et al. 1999). Tissues were suspended in 0.4 ml of cold buffer A [10 mmol/l HEPES, pH 7.9, 10 mmol/l KCl, 0.1 mmol/l EDTA, 0.1 mmol/l EGTA, 1 mmol/l 1,4-dithiothreitol (DTT), and 0.5 mmol/l PMSF] by gentle pipetting. Twenty-five microliters of a 10% Igepal CA-630 was added, and homogenates after vortexing were centrifuged ($10,000 \times g$ for 30 s). The nuclear pellet was resuspended in 50 μ l of ice-cold buffer C (20 mmol/l HEPES, pH 7.9, 0.4 mol/l NaCl, 1 mmol/l EDTA, 1 mmol/l EGTA, 1 mmol/l DTT, and 1 mmol/l PMSF), and the tube was vigorously rocked at 4°C for 15 min on a shaking platform. The nuclear extract was centrifuged at 4°C ($15,000 \times g$ for 5 min), and the supernatant was frozen at -70°C . The protein concentrations were measured using the BCA protein assay, with bovine serum albumin as a standard (Pierce, IL). NF- κ B (p65) protein was estimated in the

nuclear extracts of all by ELISA following the standard manufacturer's protocol (TransAM transcription assay kit, CA, USA).

RNA extraction and cDNA synthesis

RNA was isolated from mice kidney tissues as previously described (Chen et al. 2000, 2003a). RNA concentration was assessed on a spectrophotometer (Gene Quant, Pharmacia Biotech, USA). First-strand cDNA was made by using Superscript-II (Invitrogen, Burlington, ON, Canada) system. The resulting products were stored at -20°C .

Real time RT-PCR

Real time RT-PCR was performed in LightCyclerTM (Roche Diagnostics Canada, Laval, Quebec, Canada) to quantify the mRNA expression of heme oxygenase-1 (HO-1), fibronectin (FN), its extracellular B positive splice variant (EDB⁺ FN), collagen α -1(IV), vascular endothelial growth factor (VEGF), endothelin-1 (ET-1) and transforming growth factor- β 1 (TGF- β 1) with some modifications (Chen et al. 2000, 2003b; Evans et al. 2000; Lamy et al. 2011). For a final reaction volume of 20 μl , the following reagents were added: 10 μl of SYBR (Sigma–Aldrich), 1.6 μl of MgCl_2 , 1 μl forward/reverse primer (Table 1), 4.4 μl of H_2O , and 2 μl of cDNA. To optimize the amplification of the genes, melting curve analysis (MCA) was used to determine the melting temperature (T_m) of specific products and primer dimers. According to the T_m value of specific products for respective genes, an additional step (signal acquisition step, $2-3^{\circ}\text{C}$ below T_m) was added after the elongation phase of RT-PCR. This additional step in the PCR reactions allowed for signal acquisition from specific target products (Chen et al. 2003a). The data were normalized to 18sRNA to account for differences in reverse transcription efficiencies and amount of template in the reaction mixtures and expressed relative to control groups.

Protein extraction and ELISA

Kidney tissues were washed with cold phosphate buffered saline (PBS), homogenized and treated with lysis buffer (50 mmol/l HEPES, pH 7.6, 150 mmol/l NaCl, 50 $\mu\text{mol/l}$ NaF, 2 mmol/l EDTA, 1 mmol/l sodium vanadate, 1% NP-40, and 2 mmol/l phenyl methyl sulphonyl fluoride). The total protein concentration was measured using BCATM protein assay kit (Pierce, Rockford, IL, USA) according to the manufacturer's instructions. The concentrations of all samples

were adjusted to 500 ng/ μl before performing ELISA. We performed FN (Kamiya Biomedicals, WA, USA) and VEGF (Invitrogen, Canada) protein measurement by ELISA following the manufacturer's protocols.

Histological analysis

Formalin-fixed tissues embedded in paraffin were sectioned at 5 μm thickness on positively charged slides. The sections were stained with hematoxylin and eosin and periodic acid-schiff stain (PAS).

Statistical analysis

The data are expressed as mean $\pm$ standard error of the mean. Statistical significance was determined by analysis of variance (ANOVA) followed by the Bonferroni–Dunn test. Differences were considered to be statistically significant at values of $p < 0.05$.

Results

Phytochemical characteristics of ginseng extract

Characterization of ginseng extract and the corresponding HPLC analysis data has been previously published (Sen et al. 2011; Azikey et al. 2011). Overall, the extract showed 28.25% ginsenosides content. Rb1 and Re were two predominant ginsenosides with Rb1/Re ratio of 1.8%, which is a characteristic feature of American ginseng. The Rb1/Rg1 ratio was found to be 56.74%.

Clinical monitoring

Dysmetabolic state in diabetic animals was assessed by monitoring the body weight, blood glucose, and glycated hemoglobin (GlycoHb) levels. In the type 1 diabetic model, both the DM1 (2 months) and DM2 (4 months) groups showed significant diminution of body weight enhancement of blood glucose and GlycoHb levels as compared to the age and sex-matched (CO1 and CO2) groups, respectively (Table 2). It should also be mentioned that reduction of body weight and increase in GlycoHb levels were pronounced in 4 months compared to the 2 months group. Ginseng treatment caused significant improvement of body weight, reduction of blood glucose and GlycoHb levels both after 2 months and 4 months of follow-up (DM1-GIN and DM2-GIN) (Table 2).

On the other hand, untreated type 2 diabetic mice (db1 and db2 groups) showed significant increase of body weight, blood glucose, and GlycoHb levels compared to respective control groups (Table 3). Ginseng treatment caused a significant reduction of the body weight, blood glucose and GlycoHb levels in all the treated diabetic animals (db1-GIN and db2-GIN) (Table 3). Measurements of insulin levels have been reported to be a valuable index of insulin secretion (Doda 1996). Plasma insulin levels in db/db mice were found to be increased compared to the controls (Table 4). Ginseng

Table 1
Primer sequences for real time RT-PCR.

Gene	Sequence (5' $\rightarrow$ 3')
18sRNA	GTAACCCGTTGAACCCATT CCATCCAACGGTAGTAGCG
HO-1	TGAACACTCTGGAGATGACA AACAGGAAGCTGAGAGTGAG
Total FN	CGGTAGGACCTTCTATTCTT GATACATGACCCCTTCATTG
EDB ⁺ FN	AGTTAGTTGCGGCAGGAGAA CCGCCATTAATGAGAGTGAT
Collagen- α 1(IV)	ACAGGCACAAGTTAGGAAA ATCTCTTCTCTCCCAAG
VEGF	TTACTGCTGTACCTCCACC ACAGGACGGCTTCAAGATC
ET-1	TTAGCAAGACCATCTGTGTG GAGTTTCTCCCTGAAATGTG
TGF β 1	TGACGTCACTGGAGTTGTACG GGTTCATGTCATGGATGGTGC

Table 2
Body weights, blood glucose and GlycoHb levels in type 1 diabetic model with or without ginseng treatment.

Groups	Body wt. (g)	Blood glucose (mmol/l)	GlycoHb (%)
CO1	34 $\pm$ 2.5	8.86 $\pm$ 2.7	6.23 $\pm$ 0.8
DM1	23.5 $\pm$ 2.12 [*]	27.4 $\pm$ 8.5 [*]	8.7 $\pm$ 0.9 [*]
DM1-GIN	27.2 $\pm$ 1.7	15.28 $\pm$ 4.5	6.55 $\pm$ 0.5
CO2	28 $\pm$ 5.5	7.5 $\pm$ 1.52	6.0 $\pm$ 2.4
DM2	19.1 $\pm$ 1.2 [*]	25.96 $\pm$ 5.72 [*]	10.5 $\pm$ 2.10 [*]
DM2-GIN	29.6 $\pm$ 3.1	7.26 $\pm$ 2.5	6.34 $\pm$ 0.7

^{*} $p < 0.05$ vs. corresponding controls and ginseng treated groups.

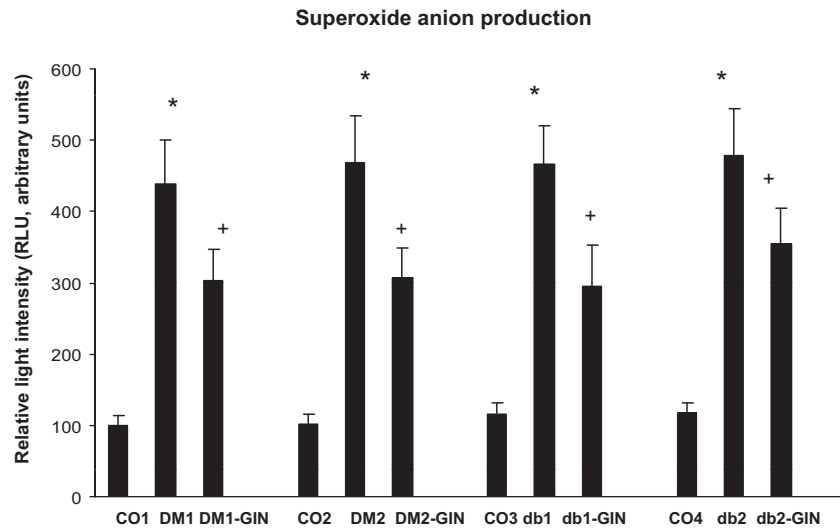


Fig. 1. Figure showing the effects of ginseng (GIN) on superoxide anion generation from the kidneys of type 1 (DM1 and DM2) and type 2 (db1 and db2) diabetic mice after 2 (DM1 or db1) or 4 months (DM2 or db2) of follow-up. Diabetes caused enhanced superoxide anion generation as evidenced by increased chemiluminescence (arbitrary units) in the mice kidneys compared to the controls (CO). Ginseng treatment prevented diabetes-induced superoxide generation [$*p < 0.05$ vs. respective controls, $+p < 0.05$ vs. DM or db, $n = 5/\text{group}$].

treatment resulted in diminution of plasma insulin levels in the db/db mice (Table 4). Inhibition of insulin after ginseng administration, suggests ginseng affects insulin resistance. In keeping with this notion, HOMA-IR indices of the treatment group were also reduced compared to those of untreated db/db mice (Table 4).

Ginseng prevents oxidative stress in the kidneys of diabetic mice

As oxidative stress is an important pathogenetic mechanism leading to chronic diabetic complications, we investigated oxidative stress using several parameters. Diabetic animals (DM1,

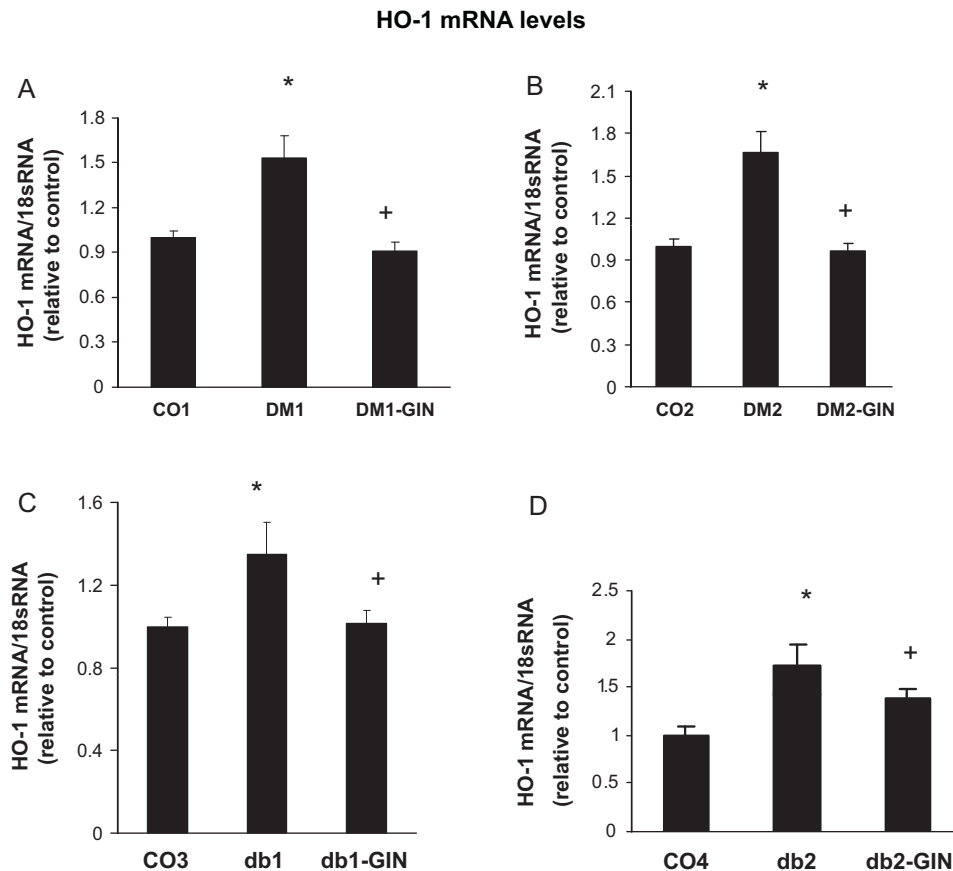


Fig. 2. Real-time RT-PCR analysis of HO-1, a molecular marker of oxidative stress, showed that diabetes-induced increased mRNA is prevented with ginseng in (A) DM1-GIN (2 months follow-up) or (B) DM2-GIN (4 months follow-up) compared to the respective diabetic groups (DM1 or DM2) and (C) db1-GIN (2 months follow-up) or (D) db2-GIN (4 months follow-up) compared to the db1 or db2 groups. Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO), $+p < 0.05$ vs. diabetics (DM or db), $n = 5/\text{group}$, HO-1: heme oxygenase-1].

Table 3

Body weights, blood glucose and GlycoHb levels in type 2 diabetic model with or without ginseng treatment.

Groups	Body wt. (g)	Blood glucose (mmol/l)	GlycoHb (%)
CO3	32 ± 2.5	8.86 ± 2.7	6.23 ± 0.8
db1	44.4 ± 2.5*	29.08 ± 2.5*	11.05 ± 3.4*
db1-GIN	37.5 ± 2.9	19.07 ± 4.5	7.5 ± 2.1
CO4	34 ± 3.5	8.86 ± 2.7	6.23 ± 0.8
db2	48.4 ± 2.06*	30.05 ± 2.5*	11.05 ± 2.10*
db2-GIN	32.6 ± 3.5	15.5 ± 3.9	6.5 ± 2.3

* $p < 0.05$ vs. corresponding controls and ginseng treated groups.

Table 4

Insulin levels and HOMA-IR index in type 2 diabetic model with or without ginseng administration.

Groups	Insulin (mU/l)	HOMA-IR
CO3	25.13 ± 2.04	10.7 ± 3.7
db1	57.97 ± 1.5*	72.4 ± 5.5*
db1 GIN	21.05 ± 3.4	16.4 ± 2.2
CO4	31.06 ± 3.03	14.07 ± 7.82
db2	60.75 ± 0.99*	81 ± 1.32*
db2-GIN	50.43 ± 3.8	35.7 ± 6.3

* $p < 0.05$ vs. corresponding controls and ginseng treated groups.

DM2, db1 and db2) exhibited significantly higher superoxide anion generation in the kidney tissues (Fig. 1) as measured by luminol–EDTA–Fe enhanced chemiluminescence compared to the control groups. Diabetic animals further exhibited increased HO-1 transcript levels, a molecular marker of oxidative stress (Fig. 2A–D). Ginseng treatment for 2 and 4 months reduced the HO-1 upregulation in all of the ginseng treated diabetic animals (Fig. 2A–D).

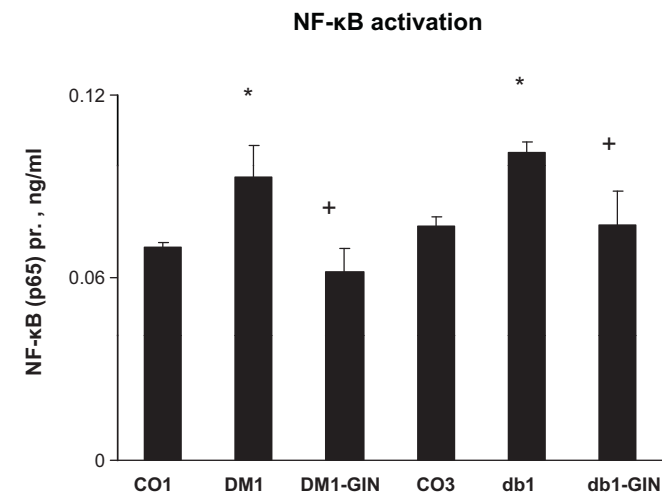


Fig. 3. Diabetes caused NF-κB (p65) activation in renal nuclei of the kidneys of DM1 (2 months) or db1 (2 months) compared to the respective controls (CO1 for type 1 or CO3 for type 2). Ginseng treatment resulted in significant prevention of NF-κB activation in DM1-GIN (2 months follow-up) or db1-GIN (2 months follow-up) compared to the respective diabetic groups (DM1 or db1). Data expressed as a ratio to 18sRNA normalized to respective controls [* $p < 0.05$ vs. controls (CO1 or CO3), + $p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group].

Ginseng prevents NF-κB activation in the renal tissues of diabetic mice

Oxidative stress-mediated NF-κB activation plays an important role in the transcription of genes responsible for glucose-induced tissue damage (Han et al. 2005). This prompted us to measure NF-κB (p65) levels in the nuclear proteins extracted from the kidneys. As this is an initiating event, we carried out the assay only after

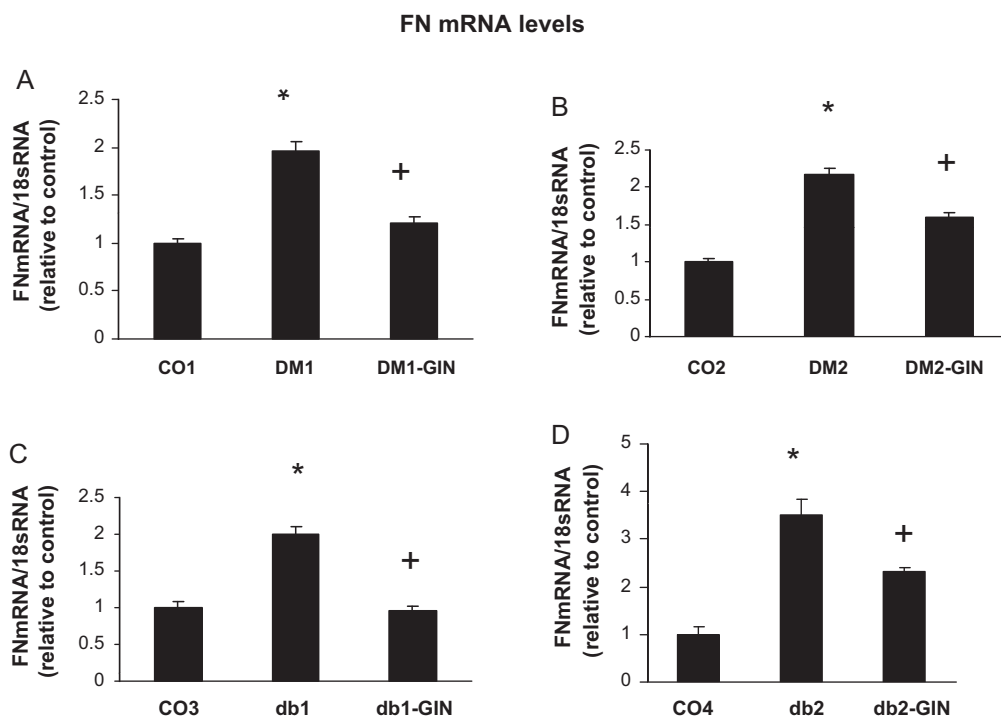


Fig. 4. Real-time RT-PCR analysis of FN showed that diabetes-induced increased mRNA is prevented with ginseng in (A) DM1-GIN (2 months follow-up) or (B) DM2-GIN (4 months follow-up) compared to the respective diabetics (DM1 or DM2) and (C) db1-GIN (2 months follow-up) or (D) db2-GIN (4 months follow-up) compared to the db1 or db2 groups. Data expressed as a ratio to 18sRNA normalized to respective controls [* $p < 0.05$ vs. controls (CO), + $p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group, FN: fibronectin].

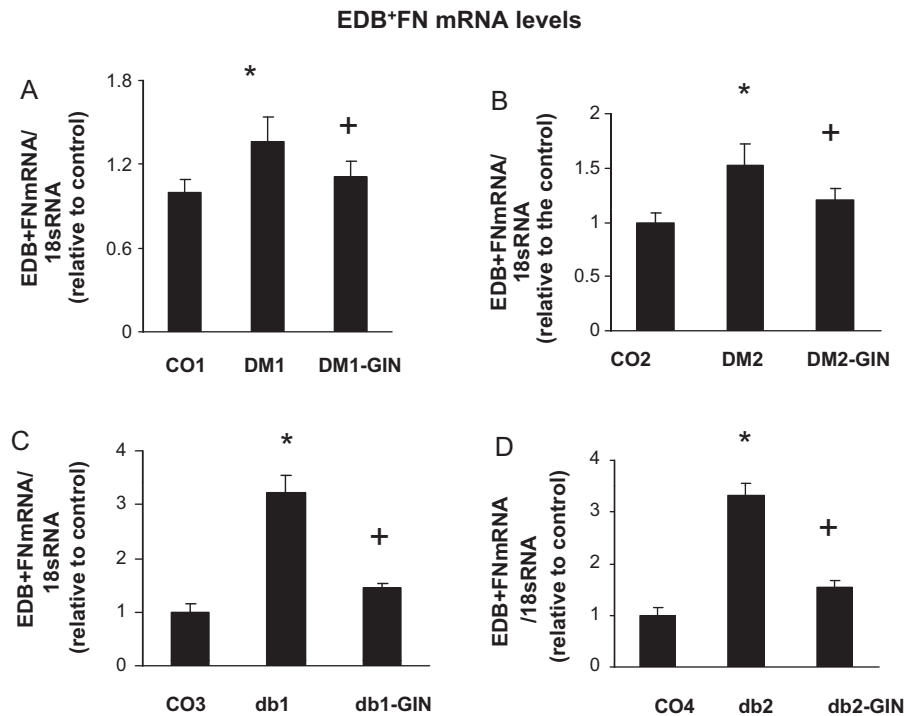


Fig. 5. Real-time RT-PCR analysis of EDB*FN (extradomain B positive splice variant of FN) showed that diabetes-induced increased mRNA upregulation in the kidneys is prevented by ginseng in (A) DM1-GIN (2 months follow-up) or (B) DM2-GIN (4 months follow-up) compared to the DM1 or DM2 groups and (C) db1-GIN (2 months follow-up) or (D) db2-GIN (4 months follow-up) compared to the db1 or db2 groups. Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO), $+p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group, EDB*FN: EDB-fibronectin].

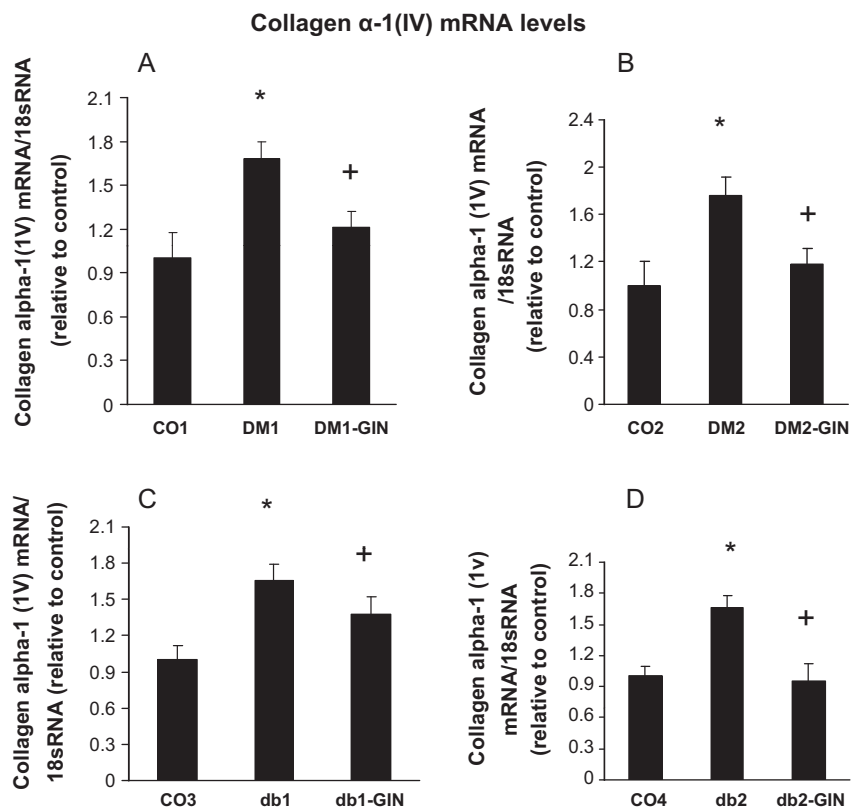


Fig. 6. Quantitative RT-PCR analysis of collagen α -1(IV) mRNA showed significant enhancement in diabetic mice (A) DM1 (2 months) or (B) DM2 (4 months) in comparison to the controls (CO1 or CO2). Similarly, collagen α -1(IV) mRNA upregulation occurred in (C) db1 (2 months) and (D) db2 (4 months) compared to the respective controls (CO3 or CO4). Treatment of the diabetic mice with ginseng for 2 or 4 months caused significant prevention of such upregulation in DM1-GIN (2 months follow-up) or DM2-GIN (4 months follow-up) compared to the DM1 or DM2 groups and db1-GIN (2 months follow-up) or db2-GIN (4 months follow-up) compared to respective db1 or db2 groups. Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO), $+p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group].

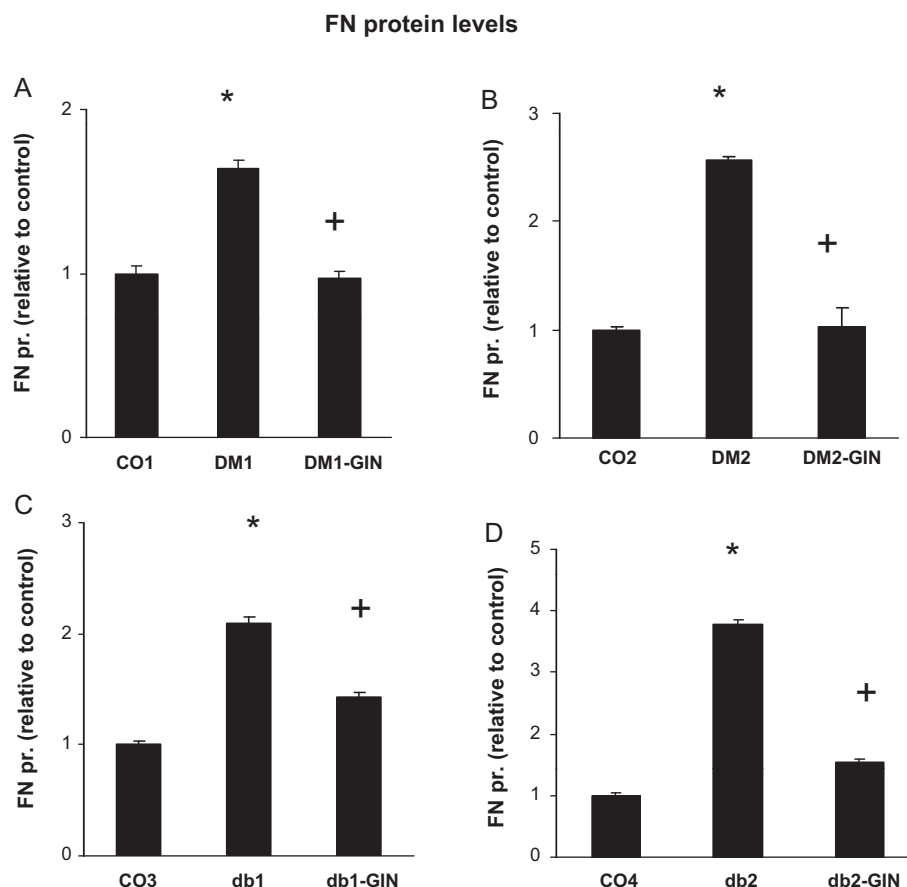


Fig. 7. Diabetes caused upregulation of FN protein (FN pr.) production in the kidneys of (A) DM1 compared to CO1, (B) DM2 compared to CO2, (C) db1 compared to CO3 and (D) db2 compared to CO4. Ginseng treatment resulted in prevention of such up regulation of FN pr. production in kidneys of DM1-GIN (2 months follow-up), DM2-GIN (4 months follow-up), and db1-GIN (2 months follow-up) and db2-GIN (4 months follow-up) compared to the respective control groups. Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO1 or CO3), $+p < 0.05$ vs. diabetics (DM or db), $n = 5/\text{group}$, FN: fibronectin].

2 months. NF- κ B (p65) level was found to be increased in all the diabetic mice compared to the controls (Fig. 3). Ginseng treatment for 2 months prevented NF- κ B activation in the diabetic animals.

Ginseng prevents upregulation of ECM mRNAs and protein production in kidneys of diabetic mice

Since glucose causes upregulation of ECM proteins in diabetes, we examined the effects of ginseng on FN, EDB⁺FN and collagen α -1(IV) mRNA production in the kidney tissues of all groups of mice. EDB⁺FN, a splice variant of FN is upregulated in diabetes (Castellani et al. 1994). STZ-induced diabetes caused upregulation of FN (Fig. 4A and B), EDB⁺FN (Fig. 5A and B) and collagen α -1(IV) (Fig. 6A and B) mRNAs in DM1 and DM2 groups in comparison to the respective controls (CO1 and CO2). Similarly, FN (Fig. 4C and D), EDB⁺FN (Fig. 5C and D), and collagen α -1(IV) (Fig. 6C and D) mRNA overproduction was also observed in db1 and db2 groups compared to the respective control groups (CO3 and CO4). Treatment of the diabetic mice with the ginseng extract significantly prevented such upregulations at both time points (Figs. 4–6). We then examined the effect of ginseng on FN protein production in kidney tissues. Diabetes caused significant increase of FN protein production in kidneys of type 1 (Fig. 7A and B) and type 2 (Fig. 7C and D) diabetic mice compared to those of respective controls. Treatment of the diabetic mice with ginseng for two and four months caused significant prevention of the enhancement of the aforementioned protein.

Ginseng prevents upregulation of vasoactive factor mRNA levels and protein production in kidneys of diabetic mice

Hyperglycemia also leads to upregulation of vasoactive factors, which prompted us to further investigate the effects of ginseng on the production of such factors in the kidneys. We selected several such factors namely, VEGF, ET-1, and TGF- β 1, as they are known to be upregulated in several chronic diabetic complications, including diabetic nephropathy (Wang et al. 2005). Furthermore, both TGF- β 1 and ET-1 are known to regulate ECM matrix protein production (Oh et al. 1998). VEGF (Fig. 8A–D), ET-1 (Fig. 9A–D), and TGF- β 1 (Fig. 10A–D) mRNA levels increased in all untreated diabetic groups in comparison to the corresponding controls. We also investigated the VEGF protein production in the 2 months and 4 months groups. VEGF protein levels (Fig. 11A–D) were significantly increased in the 2 and 4 months diabetic groups compared to the respective controls. All such elevated mRNA and protein levels in the kidneys of diabetic animals were prevented by ginseng treatment (Figs. 8–11).

Ginseng prevented the structural damage in the kidneys occurring due to diabetes

We then investigated whether these molecular changes are reflected in the structural changes at the level of the organ. Hence, histological analyses of the kidneys were performed. PAS stains revealed mesangial expansion in the kidneys of diabetic mice

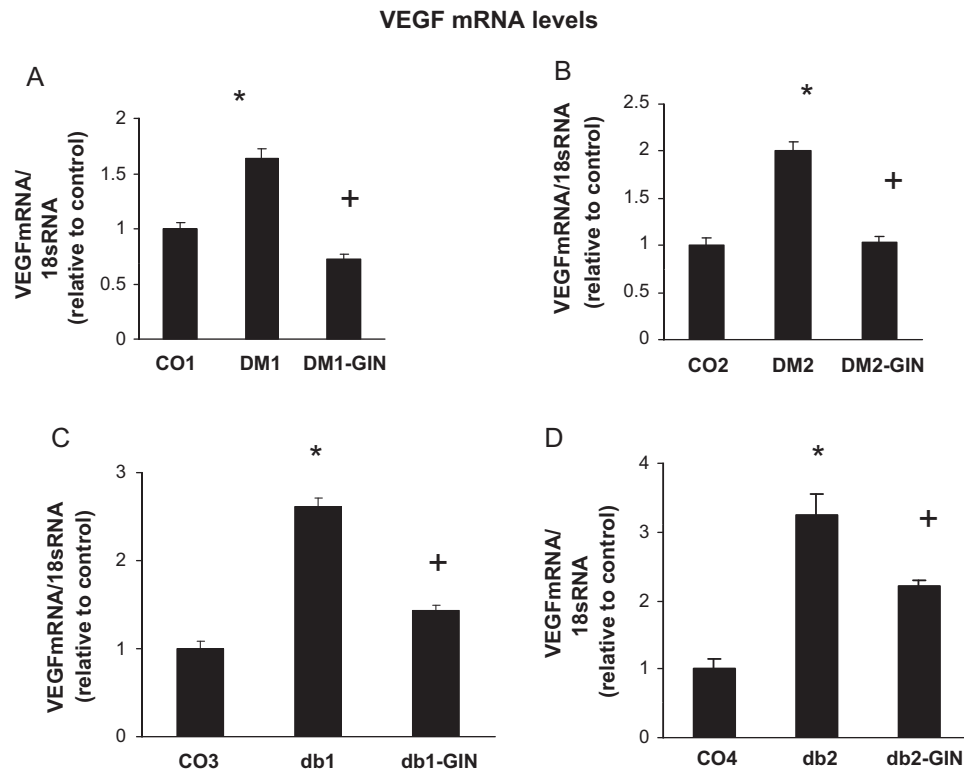


Fig. 8. Real-time RT-PCR analysis of VEGF showed that diabetes-induced increased mRNA is prevented with ginseng in (A) DM1-GIN (2 months follow-up) or (B) DM2-GIN (4 months follow-up) compared to the DM1 or DM2 groups and (C) db1-GIN (2 months follow-up) or (D) db2-GIN (4 months follow-up) compared to the db1 or db2 groups. Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO), $+p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group, VEGF: vascular endothelial growth factor].

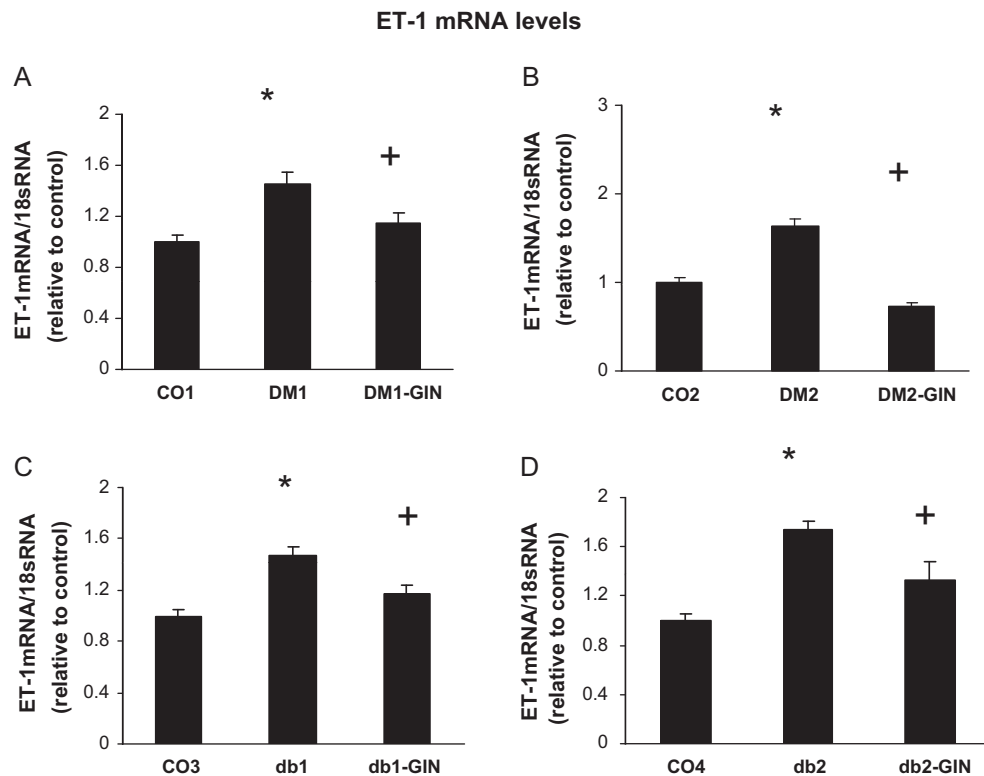


Fig. 9. Real-time RT-PCR analysis of ET-1 showed that diabetes-induced increased mRNA is prevented with ginseng treatment in (A) DM1-GIN (2 months follow-up) or (B) DM2-GIN (4 months follow-up) compared to the respective diabetic groups (DM1 or DM2) and (C) db1-GIN (2 months follow-up) or (D) db2-GIN (4 months follow-up) compared to the diabetic mice (db1 or db2). Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO), $+p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group, ET-1: endothelin-1].

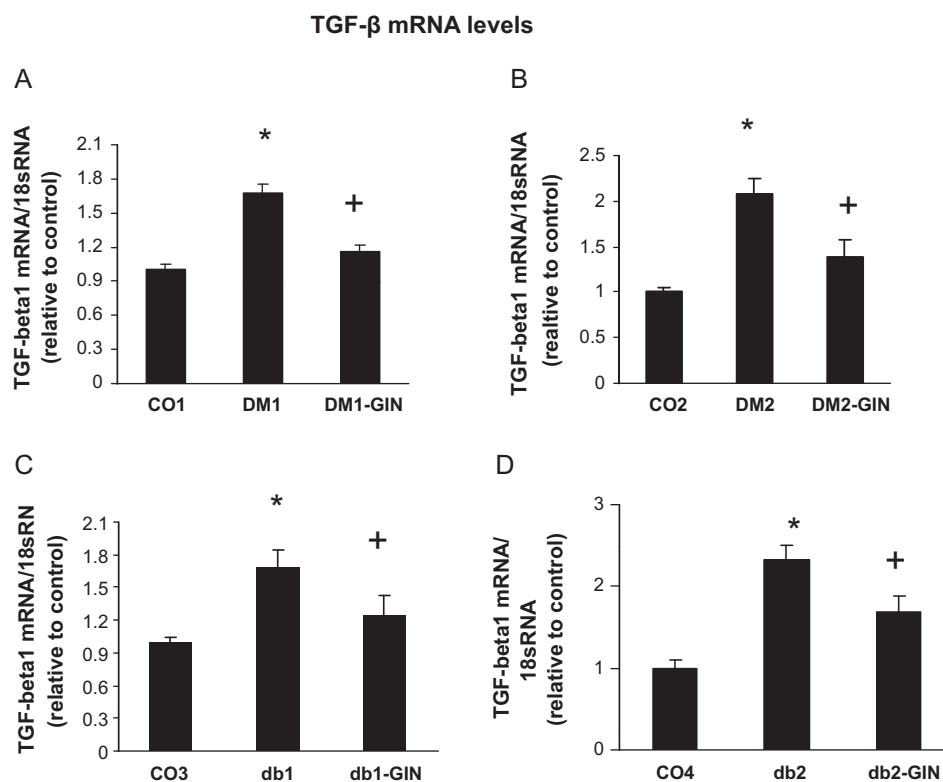


Fig. 10. Diabetes induced TGF-β1 mRNA upregulation was observed in the diabetic groups that found to be counteracted by ginseng treatment in (A) DM1-GIN (2 months follow-up) or (B) DM2-GIN (4 months follow-up) and (C) db1-GIN (2 months follow-up) or (D) db2-GIN (4 months follow-up) in comparison to the respective diabetic groups (DM1 or DM2 for type1 and db1 or db2 for type 2). Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO), $+p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group, TGF-β1: transforming growth factor-β1].

(Fig. 12B and E). Such changes were not evident in the controls (Fig. 12A and D) and were prevented in the diabetic mice following ginseng treatment (Fig. 12C and F).

Ginseng prevented renal functional alterations in diabetes

To determine whether along with biochemical and structural alterations, ginseng can prevent clinically relevant renal function abnormalities, we investigated urinary albumin levels and serum creatinine levels. In both diabetic models, significant increase in these two parameters was observed (Table 5). Oral administration of ginseng to the diabetic animals prevented the renal function abnormalities.

Table 5

Urinary albumin and serum creatinine levels in type 1 and type 2 diabetic models with or without ginseng treatment.

Groups	Urinary albumin (μg/ml)	Serum creatinine (mg/dl)
CO1	0.128 ± 0.01	2.71 ± 0.146
DM1	0.182 ± 0.02*	3.95 ± 0.325*
DM1-GIN	0.138 ± 0.007	2.59 ± 0.343
CO2	0.126 ± 0.02	2.59 ± 0.1
DM2	0.197 ± 0.019*	3.31 ± 0.45*
DM2-GIN	0.134 ± 0.009	2.47 ± 0.47
CO3	0.128 ± 0.01	2.71 ± 0.146
db1	0.204 ± 0.02*	3.33 ± 0.03*
db1-GIN	0.139 ± 0.02	2.06 ± 0.5
CO4	0.128 ± 0.01	2.59 ± 0.1
db2	0.224 ± 0.05*	3.73 ± 0.26*
db2-GIN	0.130 ± 0.09	2.08 ± 0.06

* $p < 0.05$ vs. corresponding controls and ginseng treated groups.

Discussion

In the present study, we demonstrated the preventive role of ginseng therapy on DN. We used a variety of approaches ranging from molecular to structural and functional studies. We employed animal models of both type 1 and type 2 diabetes. Both the STZ-induced diabetic and db/db mice exhibited dysmetabolic state namely body wt. abnormalities, hyperglycemia, abnormalities of insulin levels and HOMA-IR index (in db/db mice). Ginseng treatment improved such parameters, indicating that one of the mechanisms of its function may be mediated through improving dysmetabolic state in diabetes. Karaca et al. (2010) have reported about the increased plasma insulin and C-peptide levels following treatment with American ginseng with or without green tea. Analysis of the composition of the alcoholic extract of ginseng used in the present study showed presence of the major ginsenosides Rg1, Re, Rb1, Rc, Rb2 and Rd with the predominance of the Rb1 and Re ginsenosides, a characteristic feature of American ginseng. Xiong et al. (2010) reported about the antihyperglycemic and antiobesity properties of the ginsenoside, Rb1 in rats. However, from a practical standpoint use of root extract is advantageous as it combines effects of multiple ginsenosides. In previous studies, other ginsenosides such as Rh2 also demonstrated glucose lowering effect in STZ-induced rats (Lee et al. 2006). Furthermore, in keeping with our findings, previous studies have reported improvement of plasma insulin, C-peptide levels and HOMA-IR index in db/db mice after treatment with compound K, the major intestinal metabolite of ginsenosides alone or along with metformin (Yoon et al. 2007).

However, in our study although hyperglycemia was only partially corrected following ginseng treatment, major improvement

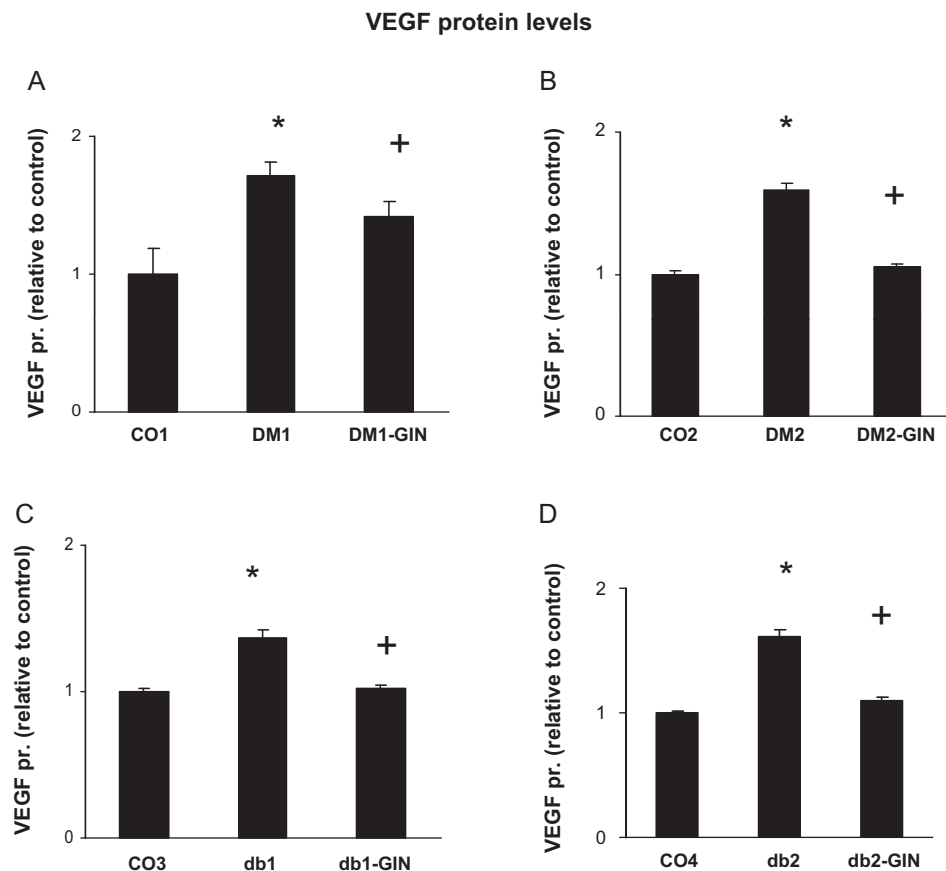


Fig. 11. Diabetes caused upregulation of VEGF protein (VEGF pr.) production in the kidneys of (A) DM1 compared to CO1, (B) DM2 compared to CO2, (C) db1 compared to CO3 and (D) db2 compared to CO4. Ginseng treatment resulted in prevention of such upregulation of VEGF pr. production in kidneys of DM1-GIN (2 months follow-up), DM2-GIN (4 months follow-up), and db1-GIN (2 months follow-up) and db2-GIN (4 months follow-up) compared to the respective control groups. Data expressed as a ratio to 18sRNA normalized to respective controls [$*p < 0.05$ vs. controls (CO1 or CO3), $+p < 0.05$ vs. diabetics (DM or db), $n = 5$ /group, VEGF: vascular endothelial growth factor].

of renal biochemical, structural and functional parameters were observed. Such findings suggest additional effects of ginseng are possibly mediated through modulation of oxidative stress. We have previously demonstrated such direct effects on endothelial cells (Sen et al. 2011). Sustained hyperglycemia is an important causal factor in mediating the development and progression of diabetic kidney disease (Ha and Kim 1999; Roy et al. 2010; Mistry et al. 2009). Multiple biochemical mechanisms have emerged to explain the adverse effects of hyperglycemia among which oxidative stress is prominent (King and Loeken 2004; Madonna and Caterina 2011). Here, we demonstrated diabetes-induced upregulation of HO-1 mRNA expression; enhanced superoxide anion generation and NF- κ B activation were prevented by ginseng treatment. Glomerular HO-1 increment secondary to diabetes-induced oxidative stress has been reported in diabetic nephropathy (Song et al. 2007; Chiu et al. 2009). Reduced antioxidant defense responses may also give rise to increased oxidative stress in diabetic condition (Beal 2002; Arai et al. 1987; Yan and Harding 1997; Voces et al. 1999). Ginseng is known to induce the transcription of Cu,Zn-superoxide dismutase gene (SOD1), which is one of the major antioxidant enzymes. It has been also reported that *Panax ginseng* (Asian ginseng) administration increased the activity of the antioxidant enzymes superoxide dismutase (SOD) and glutathione peroxidase (GPX) in rats (Voces et al. 1999). These findings suggest that some of the observed effects of ginseng are possibly mediated through its antioxidant property.

ROS stimulate the expression of various transcription factors such as NF- κ B (Scheck et al. 1992). We demonstrated that NF- κ B (p65) activation in the kidneys in diabetes, which was prevented by ginseng treatment. This may denote one of the mechanisms responsible for anti-oxidant and anti-inflammatory property of ginseng. We and others have previously demonstrated that NF- κ B activation is instrumental in the production of vasoactive factors and ECM proteins (Xu et al. 2008; Sharma et al. 1996). Here, we demonstrated upregulation of multiple ECM proteins and vasoactive factors as well as mesangial expansion in the kidneys of diabetic mice, which were prevented by ginseng therapy. In addition, ginseng treatment prevented TGF- β 1 production. TGF- β 1 is known to mediate glucose-induced FN synthesis in diabetes (Sharma et al. 1996; Wang et al. 2005). Vasoactive factors such as ET-1 also act as an upstream mediator of FN expression in diabetes through transcription factor NF- κ B activation (Chen et al. 2003a,b). Although there are no known studies regarding such parameters in the kidneys, Yuan and Chung (2010) have shown the protective effect of ginseng against STZ-induced pancreatic β -cell damage by inhibiting the NF- κ B activity. Finally, effects of ginseng were reflected in the prevention of diabetes induced renal structural and functional abnormalities. We have shown that ginseng prevented enhanced albuminuria and alterations of serum creatinine levels as well as mesangial matrix expansion, well established markers of renal damage in diabetes (Steffes et al. 1989). This is the first report showing the preventing effects of American ginseng in diabetic nephropathy in both type 1 and type 2 diabetes.

Kidney histology

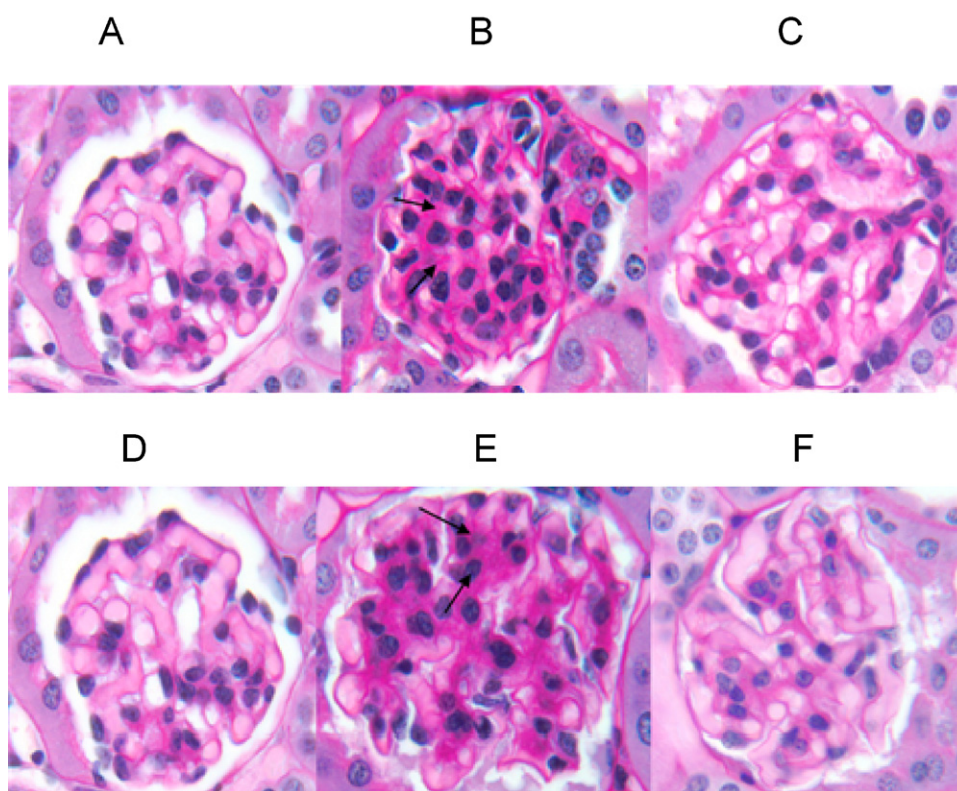


Fig. 12. Histological analysis utilizing PAS stain showing that diabetes-induced mesangial expansion found in (B) DM1 (2 months diabetic) or (E) db1 (2 months diabetic) compared with respective controls (A) CO1 or (B) CO3 was prevented with ginseng treated (C) DM1-GIN (2 months follow-up) or (D) db1-GIN (2 months follow-up). Original magnification 40× for all micrographs.

Conclusion

The present study demonstrates that North American ginseng is effective in the prevention of diabetic nephropathy both in type 1 and type 2 animal models through a combination of mechanisms namely, its antihyperglycemic and antioxidant effects. Thus ginseng may potentially lend itself as a relatively safe and inexpensive adjuvant treatment and management for diabetic nephropathy. However, it is also possible that other actions of ginsenosides unrelated to antihyperglycemic and antioxidant effects may further be responsible for the observed effects. Hence, additional pharmacokinetic studies are necessary prior to undertaking any clinical phase one studies.

Conflict of interest

No conflict to disclose.

Acknowledgement

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References

Arai, K., Maguchig, S., Fujii, S., Ishibashill, H., Oikawa, S.K., Taniguch, N., 1987. Glycation and inactivation of Cu–Zn superoxide dismutase. Identification of the in vitro glycated sites. *J. Biol. Chem.* 262, 16969–16972.

- Azike, C.G., Charpentier, P.A., Jirui, H., Hua, P., Lui, E.M.K., 2011. The yin and yang actions of North American Ginseng root in modulating the immune function of macrophages. *Chin. Med.* 6, 21–32.
- Beal, M.F., 2002. Oxidatively modified proteins in aging and disease. *Free Radic. Biol. Med.* 32, 797–803.
- Bonnefont-Rousselot, D., 2002. Glucose and reactive oxygen species. *Curr. Opin. Clin. Nutr. Metab. Care* 5, 561–568.
- Brownlee, M., 2001. Biochemistry and molecular cell biology of diabetic complications. *Nature* 414, 813–820.
- Castellani, P., Viale, G., Dorcaratto, A., Nicolo, G., Kaczmarek, J., Querze, G., Zardi, L., 1994. The fibronectin isoform containing the ED-B oncofetal domain: a marker of angiogenesis. *Int. J. Cancer* 59, 612–618.
- Chen, S., Apostolova, M.D., Cherian, M.G., Chakrabarti, S., 2000. Interaction of endothelin-1 with vasoactive factors in mediating glucose induced increased permeability in endothelial cells. *Lab. Invest.* 80, 1311–1321.
- Chen, S., Khan, Z.A., Cukiernik, M., Chakrabarti, S., 2003a. Differential activation of NF- κ B and AP-1 in increased fibronectin synthesis in target organs of diabetic complications. *Am. J. Physiol. Endocrinol. Metab.* 284, 1089–1097.
- Chen, S., Mukherjee, S., Chakraborty, C., Chakrabarti, S., 2003b. High glucose-induced, endothelin-dependent fibronectin synthesis is mediated via NF- κ B and AP-1. *Am. J. Physiol. Cell Physiol.* 284, 263–272.
- Chiu, J., Khan, Z.A., Farhangkhoe, H., Chakrabarti, S., 2009. Curcumin prevents diabetes-associated abnormalities in the kidneys by inhibiting p300 and nuclear factor- κ B. *J. Nutr.* 25, 964–972.
- Dey, L., Xie, J.T., Wang, A., Wu, J., Maleckar, S.A., Yuan, C.S., 2003. Anti-hyperglycemic effects of ginseng: comparison between root and berry. *Phytomedicine* 10, 600–605.
- Diabetes Control Complications Trial Research Group, 1993. The effect of intensive treatment of diabetes on the development of long-term complications in insulin-dependent diabetes mellitus. *N. Engl. J. Med.* 329, 977–986.
- Doda, R.F., 1996. Diabetes mellitus. In: Kaplan, L.A., Amadeo, J.P. (Eds.), *Clinical Chemistry*. Mosby Year Book, St. Louis, pp. 613–641.
- Evans, T., Deng, D.X., Chen, S., Chakrabarti, S., 2000. Endothelin receptor blockade prevents augmented extracellular matrix component mRNA expression and capillary basement membrane thickening in the retina of diabetic and galactose-fed rats. *Diabetes* 49, 662–666.

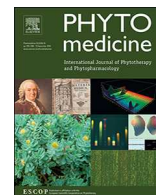
- Feng, B., Chen, S., George, B., Feng, Q., Chakrabarti, S., 2010. miR133a regulates cardiomyocyte hypertrophy in diabetes. *Diabetes Metab. Res. Rev.* 26, 40–49.
- Ha, H., Kim, K.H., 1999. Pathogenesis of diabetic nephropathy: the role of oxidative stress and protein kinase C. *Diabetes Res. Clin. Pract.* 45, 147–151.
- Han, H.J., Lee, Y.J., Park, S.H., Lee, J.H., Taub, M., 2005. High glucose-induced oxidative stress inhibits Na⁺/glucose co-transporter activity in renal proximal tubule cells. *Am. J. Physiol. Renal Physiol.* 288, 988–996.
- Howard, J., Goldberg, H.J., Catharine, I., Whiteside, C.L., Fantus, L.G., 2002. The hexosamine pathway regulates the plasminogen activator inhibitor-1 gene promoter and Sp1 transcriptional activation through protein kinase C- β 1 and - δ . *J. Biol. Chem.* 277, 33833–33841.
- Hwang, S.Y., Sohn, S.H., We, J.J., Yang, J.B., Kyung, J.S., Kwak, Y.S., Kim, S.W., Kim, S.K., 2010. Panax ginseng improves senile testicular function in rats. *J. Ginseng Res.* 34, 327–335.
- Ishii, H., Koya, D., King, G.L., 1998a. Protein kinase C activation and its role in the development of vascular complications in diabetes mellitus. *J. Mol. Med.* 76, 21–31.
- Ishii, H., Tada, H., Isogai, S., 1998b. An aldose reductase inhibitor prevents glucose-induced increase in transforming growth factor- β and protein kinase C activity in cultured human mesangial cells. *Diabetologia* 41, 362–364.
- Jung, C.H., Seog, H.M., Choi, I.W., Choi, H.D., Cho, H.Y., 2005. Effects of wild ginseng (*Panax ginseng* C.A. Meyer) leaves on lipid peroxidation levels and antioxidant enzyme activities in streptozotocin diabetic rats. *J. Ethnopharmacol.* 98, 245–250.
- Karaca, T., Yoruk, M., Yoruk, I.H., Uslu, S., 2010. Effects of extract of green tea and ginseng on pancreatic beta cells and levels of serum glucose, insulin, cholesterol and triglycerides in rats with experimentally streptozotocin-induced diabetes: a histochemical and immunohistochemical study. *J. Anim. Vet. Adv.* 9, 102–107.
- Kim, S.H., Park, S.H., 2003. Effects of Panax ginseng extract on lipid metabolism in humans. *Pharmacol. Res.* 48, 511–513.
- Kim, Y.K., Guo, Q., Packer, L., 2002. Free radical scavenging activity of red ginseng aqueous extracts. *Toxicology* 72, 149–156.
- King, G.L., Loeken, M.R., 2004. Hyperglycemia-induced oxidative stress in diabetic complications. *Histochem. Cell Biol.* 2, 332–338.
- Lamy, S., Bédard, V., Labbé, D., Sartelet, H., Barthomeuf, C., Gingras, D., Béliveau, R., 2011. The dietary flavones apigenin and luteolin impair smooth muscle cell migration and VEGF expression through inhibition of PDGFR- β . Phosphorylation. *Cancer Prev. Res.* 6, 452–459.
- Lee, F.C., 1992. Facts About Ginseng: The Elixir of Life. Hollyn International Corporation, Elizabeth, NJ.
- Lee, J.I., Ganster, R.W., Geller, D.A., Burckart, G.J., Thomson, A.W., Lu, A.L., 1998. Cyclosporine inhibits the expression of co-stimulatory molecules on in vitro-generated dendritic cells: association with reduced nuclear translocation of nuclear factor kappa B. *Transplantation* 8, 255–263.
- Lee, W.K., Kao, S.T., Liu, M.I., Cheng, J.T., 2006. Increase of insulin secretion by ginsenoside Rh2 to lower plasma glucose in wistar rats. *Clin. Exp. Pharmacol. Physiol.* 33, 27–32.
- Lee, Y.J., Pantuck, C.B., Cho, C.H., Pantuck, E.J., 1987. Effects of ginseng on the metabolism of enflurane and methoxyflurane. *Yonsei Med. J.*, 261–265.
- Liu, T.P., Liu, I.M., Cheng, J.T., 2005. Improvement of insulin resistance by Panax ginseng in fructose-rich chow-fed rats. *Horm. Metab. Res.* 37, 146–151.
- Luo, J.Z., Luo, L., 2006. American ginseng stimulates insulin production and prevents apoptosis through regulation of Uncoupling Protein-2 in cultured beta cells. *Evid. Based Complement Alternat. Med.* 3, 365–372.
- Madonna, R., Caterina, R.D., 2011. Cellular and molecular mechanisms of vascular injury in diabetes. Part II – Cellular mechanisms and therapeutic targets. *Vascul. Pharmacol.* 54, 75–79.
- Matthews, D.R., Hosker, J.P., Rudenski, A.S., Naylor, B.A., Treacher, D.F., Turner, R.C., 1985. Homeostasis model assessment: insulin resistance and beta-cell function from fasting plasma glucose and insulin concentrations in man. *Diabetologia* 28, 412–419.
- Mistry, K., Sharma, S., Andkalia, K., 2009. Possible role of glycated complement proteins in progression of diabetic nephropathy. *J. Cell Tissue Res.* 9, 1819–1897.
- Nathan, D.M., 1996. The pathology of diabetic complications: how much does the glucose hypothesis explain? *Ann. Intern. Med.* 124, 86–89.
- Oh, J.H., Ha, H., Yu, M.R., Lee, H.B., 1998. Sequential effects of high glucose on mesangial cell transforming growth factor- β 1 and fibronectin synthesis. *Kidney Int.* 54, 1872–1878.
- Ohga, S., Shikata, K., Yozai, K., Okada, S., Ogawa, D., Usui, H., Wada, J., Shikata, Y., Makino, H., 2007. Thiazolidinedione ameliorates renal injury in experimental diabetic rats through anti-inflammatory effects mediated by inhibition of NF-kappaB activation. *Am. J. Physiol. Renal Physiol.* 292, 1141–1150.
- Park, M.J., Bae, C.S., Kim, S.K., Kim, D.L., Lim, J.C., Kim, J.C., Han, J.H., Moon, J.H., Kim, K.Y., Yoon, K.C., Park, S.H., 2010. Effect of protopanaxadiol derivatives in high glucose-induced fibronectin expression in primary cultured rat mesangial cells: role of mitogen-activated protein kinases and Akt. *Arch. Pharm. Res.* 33, 151–157.
- Roy, S., Trudeau, K., Roy, S., Behl, Y., Dhar, S., Chronopoulos, A., 2010. New insights into hyperglycemia-induced molecular changes in microvascular cells. *J. Dent. Res.* 89, 116–128.
- Scheck, R., Albermann, K., Baeuerle, P.A., 1992. Nuclear factor kB: An oxidative stress-responsive transcription factor of eukaryotic cells (a review). *Free Radic. Res. Commun.* 17, 221–237.
- Schena, F.P., Gesualdo, L., 2005. Pathogenetic mechanisms of diabetic nephropathy. *J. Am. Soc. Nephrol.* 6, 30–33.
- Sen, S., Chen, S., Feng, B., Yuexiu, W., Lui, E.M.K., Chakrabarti, S., 2011. American ginseng (*Panax quinquefolius*) prevents glucose-induced oxidative stress and associated endothelial cell abnormalities. *Phytomedicine* 18, 1110–1117.
- Sharma, K., Jin, Y., Guo, J., Ziyadeh, F.N., 1996. Neutralization of TGF- β by anti-TGF- β antibody attenuates kidney hypertrophy and the enhanced extracellular matrix gene expression in STZ-induced diabetic mice. *Diabetes* 45, 522–530.
- Shin, S., Young, J.J., Park, D., Yon, J.M., Baek, I.J., Hwang, B.Y., Nam, S.Y., Young, Y.W., Kim, K.Y., Joo, S.S., Kim, Y.B., 2010. Korean red ginseng extract does not cause embryo-fetal death or abnormalities in mice. *Birth Defects Res.* 89, 78–85.
- Song, F., Qi, X., Chen, W., Jia, W., Yao, P., Nussler, A.K., Sun, X., Liu, L., 2007. Effect of *Momordica grosvenori* on oxidative stress pathways in renal mitochondria of normal and alloxan-induced diabetic mice: involvement of heme oxygenase-1. *Eur. J. Nutr.* 46, 61–69.
- Stadler, K., Jenei, V., Von-Bolcschazy, G., Somogyi, A., Jakus, J., 2003. Increased nitric oxide levels as an early sign of premature aging in diabetes. *Free Radic. Biol. Med.* 35, 1240–1251.
- Steffes, M.W., Osterby, R., Chavers, B., Mauer, S.M., 1989. Mesangial expansion as a central mechanism for loss of kidney function in diabetic patients. *Diabetes* 38, 1077–1081.
- United Kingdom Prospective Diabetic Study, 1998. *Lancet* 352, 837–853.
- Voces, J., Alvarez, A.I., Vila, L., Ferrando, A., Oliveira, C.C.D., Prieto, J.G., 1999. Effects of administration of the standardized Panax ginseng extract G115 on hepatic antioxidant function after exhaustive exercise. *Comp. Biochem. Physiol. C: Pharmacol. Toxicol. Endocrinol.* 123, 175–184.
- Wang, C., Aungi, H.H., Zhang, B., Suni, S., Li, X., Hei, H., Xie, J., Hei, T., Dui, W., Yuan, C., 2008. Chemopreventive effects of heat-processed Panax quinquefolius root on human breast cancer cells. *Anticancer Res.* 28, 2545–2552.
- Wang, W., Huang, X.R., Li, A.G., Liu, F., Li, J.H., Truong, L.D., Wang, X.J., Lan, H.Y., 2005. Signaling mechanism of TGF- β 1 in prevention of renal inflammation: role of Smad7. *J. Am. Soc. Nephrol.* 16, 1371–1383.
- WHO monographs on selected medicinal plants, 1999. vol 1. Geneva, pp. 168–183.
- Xie, J.T., McHendale, S., Yuan, C.S., 2005. Ginseng and diabetes. *Am. J. Chin. Med.* 33, 397–404.
- Xiong, Y., Ling, S., Jiu, K.J., Tso, P., Xiong, Y., Wang, G., Wood, S., Liu, M., 2010. Antiobesity and antihyperglycemic effects of ginsenoside Rb1 in rats. *Diabetes* 59, 2502–2512.
- Xu, B., Chiu, J., Feng, B., Chen, S., Chakrabarti, S., 2008. PARP activation and the alteration of vasoactive factors and extracellular matrix protein in retina and kidney in diabetes. *Diabetes Metab. Res. Rev.* 24, 404–412.
- Yan, H., Harding, J.J., 1997. Glycation-induced inactivation and loss of antigenicity of catalase and superoxide dismutase. *Biochem. J.* 328, 599–605.
- Yao, D., Vlessidis, A.G., Gou, Y., Zhou, X., Zhou, Y., Evmiridis, N.P., 2004. Chemiluminescence detection of superoxide anion release and superoxide dismutase activity: modulation effect of *Pulsatilla chinensis*. *Anal. Bioanal. Chem.* 379, 171–177.
- Yoon, S.H., Han, E.J., Sung, J.H., Chung, S.H., 2007. Anti-diabetic effects of compound K versus metformin versus compound K-metformin combination therapy in diabetic db/db mice. *Biol. Pharm. Bull.* 30, 2196–2200.
- Yuan, H.D., Chung, S.H., 2010. Fermented ginseng protects streptozotocin-induced damage in rat pancreas by inhibiting NF- κ B. *Phytother. Res.* 24, 190–195.
- Zhou, D., Brown, S.A., Yu, T., Chen, G., Barve, S., Kang, B.C., Thompson, J.S., 1999. A high dose of ionizing radiation induces tissue specific activation of nuclear factor kappa B in vivo. *Radiat. Res.* 151, 703–709.

Update

Phytomedicine

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Corrigendum

Corrigendum to: “Preventive effects of North American ginseng (*Panax quinquefolium*) on diabetic nephropathy” [Phytomedicine 19 (2012) 494–505]



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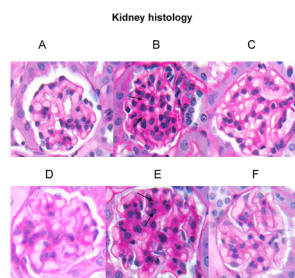
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The authors regret the errors made in the figure and figure texts of this manuscript.

The authors would like to apologise for any inconvenience caused.

Fig. 7. Diabetes caused upregulation of FN protein (FN pr.) production in the kidneys of (A) DM1 compared to CO1, (B) DM2 compared to CO2, (C) db1 compared to CO3 and (D) db2 compared to CO4. Ginseng treatment resulted in prevention of such up regulation of FN pr. production in kidneys of DM1-GIN (2 months follow-up), DM2-GIN (4 months follow-up), and db1-GIN (2 months follow-up) and db2-GIN (4 months follow-up) compared to the respective control groups. Data expressed as normalized to respective controls [$*p < 0.05$ vs. controls (CO1 or CO3), $+p < 0.05$ vs. diabetics (DM or db), $n = 5/\text{group}$, FN: fibronectin].

Fig. 11. Diabetes caused upregulation of VEGF protein (VEGF pr.) production in the kidneys of (A) DM1 compared to CO1, (B) DM2 compared to CO2, (C) db1 compared to CO3 and (D) db2 compared to CO4. Ginseng treatment resulted in prevention of such upregulation of VEGF pr. production in kidneys of DM1-GIN (2 months follow-up), DM2-GIN (4 months follow-up), and db1-GIN (2 months follow-up) and db2-GIN (4 months follow-up) compared to the respective control groups. Data expressed as normalized to respective controls [$*p < 0.05$ vs. controls (CO1 or CO3), $+p < 0.05$ vs. diabetics (DM or db), $n = 5/\text{group}$, VEGF: vascular endothelial growth factor].



group, VEGF: vascular endothelial growth factor].

Fig. 12. Histological analysis utilizing PAS stain showing that diabetes-induced mesangial expansion found in (B) DM1 (2 months diabetic) or (E) db1 (2 months diabetic) compared to respective (A,D) controls were prevented in ginseng treated (C) DM1-GIN (2 months follow-up) or (F) db1-GIN (2 months follow-up) animals. Original magnification $40\times$ for all micrographs.